Equal Pixels, Unequal Views: Field of View Limits Robust AI-Generated Image Detection

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Abstract

Detectors built on vision transformers often take a fixed-pixel window at native resolution, following the field's advice to avoid resizing. The window fixes pixel count but not field of view: a 200-pixel window sees 39% of the short side of a 512-pixel image and 5% of a 4000-pixel one. On the NTIRE 2026 Robust AI-Generated Image Detection data, the robust ROC AUC of a crop-trained DINOv2 detector falls as native image size grows. Showing the frozen detector the whole image raises fused robust AUC on the hard validation split from 0.886 to 0.963, while nine native windows at the same field of view barely help. The gain appears where the window matches the training field of view, and retraining the full models under whole-image input adds nothing (0.960), so it follows coverage rather than a train-test mismatch. Three controls argue against a resolution shortcut. A pre-registered repeat with a CLIP ViT-L/14 backbone gains 14 robust points from whole-image training, 10 of them at test time alone. On held-out benchmarks the gain holds on AIGIBench SocialRF, is confounded on CommunityAI, and reverses slightly on Synthbuster near ceiling. A declared second test read moves robust AUC by +0.062 [+0.040, +0.083] and clean AUC by -0.014 [-0.024, -0.006], a cost concentrated on large clean images. B-Free, a public detector built on native-resolution traces, does not benefit, so we limit the claim to detectors trained with our recipe. We pre-registered every gate and report two label leaks in public benchmarks.

1 Introduction

Most AI-generated image detectors must turn an image of arbitrary size into a fixed-size tensor. A long line of forensic work argues that this step should not resize: "Most of all, it is important to avoid resizing" because interpolation "may erase the subtle high-frequency traces left by the generation process" (Corvi et al. 2023). The usual answer is a fixed-pixel window at native resolution, taken once at test time or several times and averaged (Corvi et al. 2023; Li et al. 2025a; Guillaro et al. 2025; Konstantinidou, Koutlis, and Papadopoulos 2025).

A fixed-pixel window equalises pixel count. It does not equalise field of view. We define the field of view of a window of side s on an image with short side h as FOV $= s/h$. A 200-pixel window covers 39% of the short side of a 512-pixel image and 5% of a 4000-pixel one, so the larger the image, the less of it the detector sees. FixRes (Touvron et al. 2019) describes a related effect in image classification, where a change in apparent object size between training and test costs accuracy ("networks do not have built-in scale invariance"). That account, a covariate shift (Shimodaira 2000) in the input, predicts that a detector does best when the test view matches the training view. Ours does not. It does best when shown the whole image, a view it never trained on, and the gain appears even where the window already matches the training field of view. What limits it is coverage: how much of the image it sees.

We found this while building a detector for the NTIRE 2026 Robust AI-Generated Image Detection challenge (Gushchin et al. 2026). Every one of our 407k training images had a short side between 384 and 512 pixels, so the 200-pixel window we used always covered 39 to 52% of the short side. On the organisers' hard validation split it covers as little as 5%. Our detector's robust AUC fell with native image size, and three method changes we had pre-registered, aimed at the degradations, all failed their gates.

Contributions.

- A diagnosis. We show that coverage, the share of the image the detector sees, limits robust AUC for our ViT detector, and pixel count does not. Nine native windows at the same field of view barely help, while one whole-image view at the same pixel count gains 7 to 9 AUC points per branch. The gain holds where the window already matches the training field of view, so it is not a train-test mismatch (Sec. 4). A pre-registered repeat on a CLIP backbone shows the same effect, larger (Sec. 6).
- A test-time fix with controls. Presenting the whole image to the frozen, crop-trained checkpoints recovers the gain with no retraining. Our full DINOv2 models gain no more from training under whole-image input; short proxy runs gain 4 to 6 points more from it. We test and rule out a resolution shortcut three ways, and show the gain on images that received no geometric transform (Sec. 5).
- Pre-registered held-out evidence, including where it fails. Three external benchmarks give one clean pass, one confounded pass and one failure, and a declared second test

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read shows a clean-AUC cost on large images (Sec. 7).

- A pre-registered test on two public detectors. UnivFD is at chance on this data, and B-Free fails: whole-image input does not help it and costs it clean AUC (Sec. 9).
- Two label leaks in public benchmarks: a file-format and size confound in AIGIBench CommunityAI (Li et al. 2025b), and a non-monotone size leak in the resized RAISE reals commonly paired with Synthbuster (Bammey 2024) (Sec. 10).

Whole-image input is not new. The NTIRE 2026 winner resizes the whole image at inference "to preserve the global spatial context", the fifth-placed team calls its preprocessing "squish" (Gushchin et al. 2026), and AI-GenBench (Pellegrini et al. 2025) finds resize "often preferable" to multi-crop for CLIP and DINOv2 backbones. Each gives a one-line rationale: AI-GenBench notes that multi-crop uses "only partial content information", and the NTIRE entries cite global context or full image content. None separates coverage from pixel count or tests the rationale. Our contribution is that test and the evidence around it. We also show where the advice to avoid resizing still holds.

2 Related work

Crop, don't resize. Corvi et al. (2023) recommend training on native patches cropped without resizing and deciding on the whole image by fusion. SAFE (Li et al. 2025a) replaces down-sampling with a crop and reports that the resize "indeed diminishes the subtle artifact features". B-Free (Guillaro et al. 2025) trains on 504-pixel crops and averages native-resolution crops at test time. SPAI (Karageorgiou et al. 2025) learns a spectral model so that any resolution can be scored without resizing. TextureCrop (Konstantinidou, Koutlis, and Papadopoulos 2025) selects textured native crops and notes that preprocessing loss grows with native resolution. HiDA-Net (Mu et al. 2026) tiles the full image and reports 84.2% accuracy from its global view alone against 96.1% with tiles. All of these reason about high-frequency traces. SAFE's own ablation is useful context for us: its CNN scores a centre crop and the full native image about equally, so a low-level detector does not appear to depend on field of view.

Resize can win. AI-GenBench (Pellegrini et al. 2025) reports resize at least as good as five-crop for fine-tuned DINOv2 ViT-L/14 (94.24 against 93.44 AUROC) and clearly better for CLIP (92.04 against 87.66). In AIGIBench's test-time comparison (Li et al. 2025b), the one CLIP-based detector does better with resize while all five low-level detectors do better with crops. Cozzolino et al. (2024) resize the short side before a centre crop.[1] WaRPAD (Choi, Lee, and Lee 2025) rescales the whole image to a fixed square before tiling it, which holds each tile's field of view constant, although the paper motivates it differently. GlobalForge (Cui et al. 2026) reports the opposite of our test-time result on clean data: a UnivFD probe trained with centre crops loses about 9 balanced-accuracy points (67.9 to 58.7, their Fig. 2) when tested with resize. We return to this in Sec. 8.

Scale and size as confounds. Grommelt et al. (2025) show that JPEG quality and image size separate real from fake in GenImage, and note in one sentence that "detectors employing cropping to achieve a uniform input size are still subject to bias in terms of object size". Their concern is a size shortcut correlated with the label. Ours is the reverse case: size is uninformative in training and the detector still fails across sizes at test time. Rajan et al. (2025) show that random resized crops inject resampling signals when real and fake sizes differ, and BIAS-ID (Ricker, Fischer, and Quiring 2026) measures detector score shifts under such transformations. FixRes (Touvron et al. 2019) is the closest precedent. Sec. 4 shows why its train-test account does not fit our data.

3 Setup

Data. We train on 407,223 images: the NTIRE 2026 training shards, COCO and Open Images reals, and paired real and generated images from several open and commercial generators. Every training image has a short side of 384 to 512 pixels (median 512). We evaluate on the organisers' labelled splits: $\mathrm{ntire_val}$ (10,000 images, half distorted), $\mathrm{ntire_val_hard}$ (2,500, half distorted) and the public test split $\mathrm{ntire_test}$ (2,500). Robust AUC is the ROC AUC over distorted images; clean AUC is over undistorted ones, following the challenge (Gushchin et al. 2026).

Models. Two branches built on DINOv2-L/14 with registers (Oquab et al. 2024; Darcet et al. 2024), all 24 blocks fine-tuned, mean-pooled patch tokens, 0.91B parameters in total. Branch A is one tower at a 512-pixel input, trained for 2 epochs. Branch B concatenates two towers at 224 pixels, trained for 3 epochs. Each source image yields one clean view and ten degraded views drawn from a pool of 26 operations in six families, chained up to five deep, including a neural codec (Ballé et al. 2018) and a downscale applied before the window is taken. The loss is class-balanced cross-entropy plus a degradation-prediction term (weight 0.3) and clean-degraded consistency terms on the outputs (KL) and hidden features (MSE), each at unit weight. The schedule is cosine with 3% warm-up, with weight averaging (Izmailov et al. 2018) over the last quarter. On one RTX 4090 the squish-trained branches took about 19 hours (B) and 29 hours (A).

Input presentation. The *crop* policy takes a 200-pixel window at native resolution (the centre window at test time) and resizes it to the branch's input size. Images with a short side under 200 pixels are taken whole. The *squish* policy resizes the whole image to the input size, discarding aspect ratio: area interpolation on the short side down to the input size when the image is larger, then area interpolation to the square. Both policies give every image the same number of input pixels. Under squish the field of view is 1 for every image.

Fusion and selection. The two branches' logits are z-scored and combined as $wz_A + (1-w)z_B$. The z-score parameters and the weight w are fitted on $\mathrm{ntire_val}$ alone, by

[1] In their released code, https://github.com/grip-unina/ClipBased-SyntheticImageDetection.

split	n	p5	p50	p95
train	407,223	0.391	0.391	0.521
ntire_val	10,000	0.100	0.312	0.893
ntire_val_hard	2,500	0.112	0.298	0.784

Table 1: Field of view of the 200-pixel window (200/short side). Training is close to a single value. On ntire_val_hard, 86.4% of images fall outside the training band of 0.391 to 0.521 and 70.8% fall below it.

the mean of robust and clean AUC. For the squish models the choice between final and averaged weights is made the same way; for the crop models we kept the choice made earlier on both validation splits. The fitted weights were $w = 0.50$ for crop and $w = 0.55$ for squish.

Statistics. All intervals are 95% percentile intervals from 2,000 paired bootstrap resamples, stratified by label and distortion status, with the same resample indices for every model being compared.

Protocol. The test split was read once with the crop fusion (2026-10-04). We then wrote and committed a pre-registration of the squish study, with three gates and one declared second test read, before training any squish model. The gates were: G1, a three-seed proxy comparison; G2, the fused models on ntire_val_hard; and G3, each external benchmark. We amended the pre-registration four times, each time before the numbers concerned existed: after inspecting the external data for leaks, to add the aspect-ratio control (Sec. 5), to score two public detectors (Sec. 9), and to repeat the proxy comparison with a CLIP backbone (Sec. 6). We chose the squish policy after seeing its effect on ntire_val_hard, so that split is not an unbiased readout of it. The external benchmarks and the test read carry the held-out evidence. Code, the pre-registration and per-image scores will be released with the paper.

4 The diagnosis

Table 1 shows how much of each image the window covers. In training it always covers 39 to 52% of the short side. On ntire_val_hard the median is 30%, and one image in twenty gets 11% or less. Under the crop policy, fused robust AUC on ntire_val_hard falls from 0.911 for images with a short side of 80 to 336 pixels to 0.865 for images above 752 pixels. Figure 1 shows the same fall in eight equal-count bins.

To separate field of view from pixel coverage, we scored the two frozen crop-trained branches under four test-time presentations (Table 2). *grid9* averages nine 200-pixel native windows, nine times the pixels at the same field of view. *fov3* first downscales the short side to 600 pixels and then takes the nine windows, which raises the field of view to 1/3. *squish* shows the whole image once.

More pixels at the same field of view do little: grid9 gains 1.8 robust points on B and loses 0.7 on A. Gains rise with field of view: fov3 gains 1.4 and 3.4 points, and squish gains 9.2 and 7.3. Squish uses one view and the same pixel count

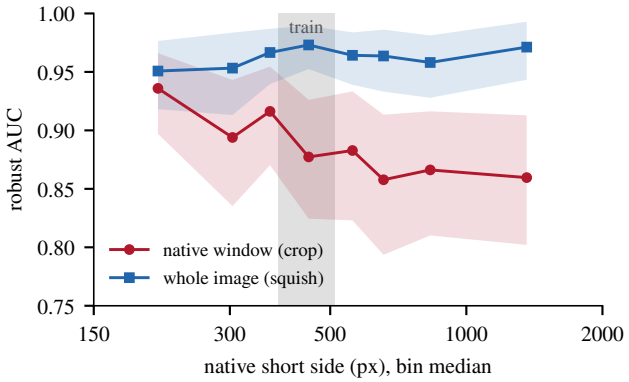


Figure 1: Fused robust AUC on the distorted rows of ntire_val_hard in eight equal-count bins of native short side, with 95% bootstrap bands. Red: the crop-trained fusion under its 200-pixel window. Blue: the fusion trained under squish. Grey: the native short sides of the training images.

branch	presentation	views	robust	clean
A (512)	centre window	1	0.8561	0.9912
	grid9	8.9	0.8495	0.9978
	fov3	8.9	0.8703	0.9977
	squish	1	**0.9482**	0.9906
B (224)	centre window	1	0.8872	0.9801
	grid9	8.9	0.9049	0.9931
	fov3	8.9	0.9215	0.9942
	squish	1	**0.9599**	0.9886

Table 2: Frozen crop-trained branches on ntire_val_hard under test-time presentations. More pixels at the same field of view (grid9) help B by 1.8 points and cost A 0.7. Raising the field of view helps in proportion.

as the centre window. On B the paired interval for the squish gain is [+0.057, +0.090]. With z-scores and weight fitted on ntire_val, the fused robust AUC rises from 0.886 to 0.963 and the true positive rate at 1% false positive rate from 0.516 to 0.752. Clean AUC moves by −0.001 on A and +0.009 on B.

Coverage, not mismatch. A train-test mismatch account, as FixRes gives for classification, predicts that the crop detector does well wherever its window matches training. Figure 1 shows otherwise. In the bin whose short sides (416 to 496 pixels) lie inside the training range, the 200-pixel window covers the same share of the image as in training, yet the crop fusion scores 0.877 against 0.973 for squish. Many of these images were downscaled from larger originals, so their content scale may still differ from training. Two further results point the same way. Squish shows the crop-trained branches a field of view of 1, which they never saw in training, and gains 7 to 9 points. Training the full branches under squish then adds nothing to that gain (Sec. 6). The detector does better the more of the image it sees, whether or not the view matches training. In short proxy runs the view match also matters: there, training under squish adds 4 to 6 points

distortions	n	short p50	A gain	B gain
photometric only	241	784	+0.090	+0.083
downscale, no crop	740	432	+0.095	+0.065
crop or aspect crop	191	320	+0.059	+0.063
perspective	78	536	+0.146	+0.140

Table 3: Squish gain in robust AUC over the centre window, within groups of distorted ntire_val_hard rows, frozen crop-trained branches. All eight 95% intervals exclude zero.

beyond the test-time change, with DINOv2 and with CLIP (Sec. 6).

5 Is it a resolution shortcut?

Squish does not hide native resolution. A small image arrives upsampled and soft, a large one downsampled and sharp, so a model could read size from the pixels. Three checks argue that the gain is not size-reading, and two more show that it does not come from the geometric distortions or from the stretch.

Size carries no label signal in training. Native short side scores an AUC of 0.528 against the label on the training set, where no image exceeds 512 pixels. A shortcut the training data does not reward is unlikely to be learned.

The gain holds at fixed size. Within equal-count deciles of native short side, where the resize factor is nearly constant, squish beats the centre window in 10 of 10 deciles on branch B. The size-weighted within-decile robust AUC rises from 0.888 to 0.964, a gain of 7.5 points against 7.3 pooled.

The gain survives band-limiting. We band-limit the squished input to 160 pixels, so native resolution can no longer be read from it. Branch B keeps all of its gain (+7.2 of +7.3 points). Branch A keeps +5.7 of +9.2. At a 512-pixel input a 160-pixel band removes real detail, so A's shortfall is the expected cost of the filter.

The gain is not only the geometric distortions. The challenge's validation splits add random crops, aspect crops and downscaling, and the hard validation and test splits also add perspective transforms; its training split has none of these (Gushchin et al. 2026). These change native size and could carry the effect. Grouping the distorted rows of ntire_val_hard by their transforms (Table 3), squish gains in every group. On rows that received only photometric transforms the gain is +9.0 points on A and +8.3 on B, with intervals clear of zero. Those rows have a median short side of 784 pixels; large native images are enough to produce the effect.

Aspect ratio. Squish stretches non-square images, which Grommelt et al. (2025) warn introduces frequency artefacts along one axis. To attribute the gain to field of view rather than to the stretch, we registered two aspect-preserving presentations before scoring them: *short-centre* (short side to the input size, then the centre square) and *letterbox* (long side to the input size, padded to a square). We predicted that each would recover at least 75% of squish's gain on both branches.

	robust gain over centre		clean vs. squish	
presentation	A	B	A	B
squish	+0.092	+0.073	–	–
short-centre	+0.085	+0.070	0.000	0.000
letterbox	+0.092	+0.069	−0.004*	−0.004*

Table 4: Differences in AUC for the aspect-preserving presentations, with the frozen crop-trained branches on ntire_val_hard and a paired bootstrap of 2000 resamples. Every robust gain has a 95% interval that excludes zero. * marks a clean difference whose interval excludes zero.

Both did (Table 4): short-centre kept 93% of the gain on A and 97% on B, letterbox 100% and 94%. Against squish, short-centre is 0.7 points lower in robust AUC on A, with an interval that excludes zero, and indistinguishable on B. Letterbox matches squish in robust AUC but loses 0.4 points of clean AUC on both branches. The stretch is therefore not what drives the gain; showing the whole short side is.

6 Does training on whole images help?

Proxy (G1). We trained matched pairs of single-tower models for one epoch on a quarter of the training set, differing only in presentation, at three seeds. On ntire_val_hard, squish beat crop by +0.083 [+0.062, +0.105], +0.093 [+0.072, +0.115] and +0.078 [+0.057, +0.102] robust AUC, with clean AUC within noise (+0.008 to +0.012, every interval spanning zero). On ntire_val the robust gain was +0.035 to +0.037. All three pre-registered conditions held, so we trained the full branches.

Full branches. Table 5 compares, per branch, the crop-trained model under its own presentation, the same frozen model under squish, and the model trained under squish. Training under squish matches the test-time change on robust AUC (A: 0.943 against 0.948; B: 0.959 against 0.960) and adds a little clean AUC. The fused squish-trained models reach 0.960 robust and 0.995 clean on ntire_val_hard, against 0.886 and 0.990 for the crop fusion: +0.074 [+0.058, +0.091] robust (G2). The size gradient is gone: across the four size buckets the squish fusion scores 0.951, 0.971, 0.965 and 0.967, and across the eight bins of Figure 1 it stays between 0.951 and 0.973.

A second backbone. Our fourth amendment repeated G1 with a CLIP ViT-L/14 tower (Radford et al. 2021) in place of DINOv2, every other setting unchanged (including ImageNet input normalisation). It added two tests: the crop-trained checkpoints shown the whole image at test time, and a mechanism check requiring the gain above 752 pixels of short side to exceed the gain below 336. Table 6 gives the means over three seeds, and every registered test passed. Training under squish beat crop by +0.144, +0.150 and +0.139 robust AUC, every interval clear of zero, with clean AUC unchanged (mean −0.001, every interval spanning zero). At test time alone the gain was +0.096, +0.111 and +0.094, again clear of zero. Both gains grow with size:

branch	model	robust	clean
A	crop-trained	0.8561	0.9913
	crop-trained, squish at test	0.9482	0.9906
	squish-trained	0.9425	0.9934
B	crop-trained	0.8873	0.9801
	crop-trained, squish at test	0.9599	0.9886
	squish-trained	0.9589	0.9898

Table 5: Per branch on ntire_val_hard. Training under squish adds nothing to robust AUC beyond presenting the whole image to the crop-trained model. Paired gains over crop-trained: A +0.092 and +0.086, B +0.073 and +0.072, all intervals clear of zero. The crop-trained rows were rescored for this table and differ from Table 2 by at most 0.0001.

backbone	model	robust	clean
DINOv2	crop-trained	0.7994	0.9255
	crop-trained, squish at test	0.8310	0.9087
	squish-trained	0.8841	0.9357
CLIP	crop-trained	0.7482	0.9655
	crop-trained, squish at test	0.8485	0.9324
	squish-trained	0.8924	0.9645

Table 6: Proxy models (one tower, one epoch on a quarter of the training set) on ntire_val_hard, mean of three seeds. The DINOv2 test-time row was not registered.

+0.206 above 752 pixels against +0.091 below 336 for training, and +0.168 against +0.057 at test time. CLIP's crop-trained proxies are weaker than DINOv2's (0.748 robust against 0.799) and gain more.

The CLIP results led us to run the same test-time comparison on the DINOv2 proxies, which we had not registered. Shown the whole image, the DINOv2 crop-trained proxies gain +0.026 to +0.038 robust AUC, every interval clear of zero, and the gain again grows with size (+0.075 above 752 pixels, +0.023 below 336). At proxy scale, training under squish adds to the test-time change for both backbones: squish-trained proxies beat the crop-trained ones shown the whole image by +0.048 to +0.057 robust and +0.022 to +0.032 clean with DINOv2, and by +0.038 to +0.048 and +0.025 to +0.040 with CLIP, every interval clear of zero. Test-time squish costs the crop-trained proxies 1 to 4 clean points, which training recovers. The full DINOv2 branches show no such gap (Table 5). They differ from the proxies in training length, number of views per image and loss terms, and we cannot say which of these closes the gap.

7 Held-out evidence

External benchmarks (G3)

We scored three external benchmarks that no model was trained or selected on, after removing any image whose content digest matched an NTIRE image: AIGIBench SocialRF and CommunityAI (Li et al. 2025b), and Synthbuster with RAISE-1k reals (Bammey 2024; Dang-Nguyen et al. 2015). All three are clean, so we report pooled AUC on the files as distributed (the gate), on every file re-encoded as JPEG at quality 95, and on every file passed through one seeded chain from our degradation pool. That last reading is our own synthesis, not the organisers', and it is not a gate.

Table 7 gives the results. On SocialRF, squish wins on the files as distributed, after JPEG re-encoding, and within size deciles. On CommunityAI the pooled gain passes the gate, but every real image is a JPEG and every fake a PNG, and native size alone separates the classes with AUC 0.916. Our first amendment required the size-matched gain on the re-encoded files to clear zero before CommunityAI could count as support. It does not ($[-0.002, +0.012]$), so we report CommunityAI as confounded. On Synthbuster squish loses by 0.001, with both fusions near ceiling. Under our degradation chains squish wins on all three sets, by 2.4 to 8.5 points.

The second test read

Table 8 gives both reads of the public test split. Read 1 used the crop fusion as frozen on 2026-09-23 and read on 2026-10-04. Read 2 used the squish fusion as frozen before any squish model was scored on test, and our pre-registration allows no third read. Robust AUC rose by +0.062 [+0.040, +0.083] and clean AUC fell by 0.014 [0.006, 0.024]. On robust AUC the squish fusion would sit between the sixth- and seventh-placed challenge entries on the open test, where the crop fusion was below all nine; we were not a challenge entry and make no ranking claim. The leading entries use far more compute, such as committees of DINOv3 models at 512 pixels on 32 A100 GPUs (Gushchin et al. 2026).

8 Where whole-image input costs

The clean loss on the test split has one source. Split by native size, clean AUC of branch A on test images with a short side above 1100 pixels falls from 0.974 under crop to 0.880 under squish (366 images). Below 1100 pixels, and on branch B at every size, the change is under 1.3 points. The real images are not the cause: under crop their mean score drifts toward "fake" as size grows, and under squish it is flat. The fake images are: their mean score under squish falls with size.

The validation splits do not show this cost. On their clean images above 1100 pixels, branch A trained under squish scores 1.000 against 0.989 for crop on ntire_val (871 images) and 0.978 against 0.981 on ntire_val_hard (295 images). Something about the test split's large generated images differs, and the split carries no generator labels, so we cannot say what. Our hypothesis is the trade-off the forensic literature predicts: a clean, large generated image keeps fine traces at native resolution, and reducing it to 512 pixels discards them. This would also reconcile our result with GlobalForge (Cui et al. 2026) and TextureCrop's CLIP results (Konstantinidou, Koutlis, and Papadopoulos 2025), both measured on clean data. Once blur, compression or downscaling has damaged the traces, the detector has to judge content, and content needs context. We have not tested this hypothesis directly.

9 Public detectors

To test whether the effect reaches beyond our detector, we registered a third amendment before scoring two pub-

	SocialRF	CommunityAI	Synthbuster
n (real/fake)	2994/3000	6000/6000	994/8996
short side p50	1366	500	833
size-label AUC	0.694	0.916	0.519
format-label AUC	0.500	1.000	0.555
crop fusion	0.9633	0.9809	0.9995
squish fusion	0.9776	0.9920	0.9984
difference (G3)	+0.014	+0.011	-0.001
95% CI	[+0.010, +0.018]	[+0.009, +0.013]	$[-0.002, -0.001]$
size-matched, JPEG 95	+0.016	+0.005	n/a
95% CI	[+0.009, +0.024]	$[-0.002, +0.012]$	
degraded (ours)	+0.085	+0.070	+0.024

Table 7: External benchmarks, fused AUC. Size-matched is the size-weighted AUC within deciles of native short side. It cannot be computed on Synthbuster because no decile holds both classes (Sec. 10).

read	robust	clean	AUC	TPR@1%
1, crop	0.7931	0.9856	0.9108	0.289
2, squish	0.8547	0.9715	0.9211	0.300

Table 8: The two reads of $\mathrm{ntire_test}$ (2,500 images). Read 1's fusion weight was fitted on both validation splits ($w = 0.45$); read 2's on $\mathrm{ntire_val}$ alone.

lic checkpoints. UnivFD (Ojha, Li, and Lee 2023) is a linear probe on CLIP ViT-L/14 features, trained on 224-pixel crops of 256-pixel images.[2] B-Free (Guillaro et al. 2025) is a DINOv2-B with registers that averages five 504-pixel token windows taken at native resolution.[3] We scored each three ways: its authors' test pipeline (native), its own training field of view (short side to its training size, then its crop), and squish at its input size. The primary test was squish minus native in robust AUC on $\mathrm{ntire_val_hard}$, with an interval that excludes zero. If no presentation reached 0.60 robust AUC, we would report the numbers and not interpret them.

UnivFD is at chance on NTIRE data under every presentation (robust AUC 0.49 to 0.51 on the hard split). That is below the floor, so it says nothing either way.

B-Free fails the primary test (Table 9). Squish changes its robust AUC on the hard split by -0.018 [-0.041, +0.004], and the change falls with image size instead of rising: -0.098 for short sides above 752 pixels, +0.014 below 336. Both resized presentations lower B-Free's clean AUC on every split, by 1.2 to 18.8 points, which is what the advice to avoid resizing predicts for a detector built on native-resolution traces (the range covers the splits in Table 9). Two results go the other way: its training field of view gains 1.9 robust points [+0.9, +2.9] on $\mathrm{ntire_val}$, and squish gains 6.3 points [+5.1, +7.5] on SocialRF under our degradation chains. As registered, we narrow the claim. Coverage limits the crop-trained detectors we fine-tuned from DINOv2 and from CLIP; we have no evidence that it limits a detector designed around native-resolution traces.

[2]Crop sizes from the released code, https://github.com/WisconsinAIVision/UniversalFakeDetect.

[3]Model size and window count from the released weights and code, https://github.com/grip-unina/B-Free.

split	native	train FOV	squish
hard-val, robust	0.636	0.631	0.618
hard-val, clean	0.909	0.753	0.722
$\mathrm{ntire_val}$, robust	0.735	0.754	0.744
$\mathrm{ntire_val}$, clean	0.920	0.848	0.825
SocialRF	0.922	0.889	0.871
degraded (ours)	0.757	0.829	0.820
CommunityAI	0.891	0.742	0.703
degraded (ours)	0.683	0.568	0.539
Synthbuster	0.998	0.987	0.976
degraded (ours)	0.896	0.893	0.892

Table 9: B-Free AUC under three presentations. Native is the authors' own five-window test pipeline. Hard-val is $\mathrm{ntire_val_hard}$.

10 Two label leaks

AIGIBench CommunityAI. All 6,000 real images are JPEG files and all 6,000 fake images are PNG files, so container format separates the classes perfectly, and native short side separates them with AUC 0.916. We found no mention of this in the paper or its public issue trackers. Results on this subset should be reported after format equalisation and within size strata.

Synthbuster with resized RAISE reals. Synthbuster ships only generated images (Bammey 2024).[4] A widely used copy of its RAISE reals, released with B-Free (Guillaro et al. 2025), was resized so that each image's area is about 1024^2 pixels;[5] we used this copy. All 994 reals then have a short side of 833 to 835 pixels, while the fakes range from 256 to 2048. The monotone size-label AUC is 0.519 because the reals sit in the middle of the fakes' range, but the rule "short side within 833 to 835" separates the classes almost perfectly. A

[4]The dataset record states that the RAISE-1k images are not included: https://zenodo.org/records/10066460.

[5]B-Free README, https://github.com/grip-unina/B-Free.

monotone leak check misses this. In the same benchmark, Adobe Firefly images saved with a .png extension are JPEG internally (Ricker, Fischer, and Quiring 2026); 997 of our fakes are JPEG.

11 What did not help

Before finding the coverage effect we pre-registered three changes aimed at the degradations, each with a written gate. Distilling degraded features toward clean ones lost about one robust point on the proxy, and under a control that gave both views the same window it still gained nothing (-0.2 points against a gate of $+1$). Conditioning the classifier on a predicted degradation embedding through FiLM tied the baseline (0.803 against 0.804). Weighting the fusion by predicted degradation severity could not clear its gate of $+0.5$ points even with an oracle (+0.4 points pooled). These are single-proxy results, but together with the probe they suggest the representation was not the bottleneck.

12 Limitations

Our full models use one backbone (DINOv2-L with registers) at two input sizes. The CLIP repeat is at proxy scale only, and both backbones went through our own fine-tuning recipe. The low-level detectors that motivate the advice to avoid resizing may not share the effect, and SAFE's results suggest they do not (Li et al. 2025a). The policy was chosen on $\mathrm{ntire_val_hard}$, so its numbers there are optimistic; the external sets and the test read are the unbiased evidence, and they are mixed. We have not shown why coverage matters on degraded images; our hypothesis, that degradations damage local traces and leave the detector to judge content spread across the image, is untested. The clean cost on large images is real and is not addressed here. Combining a whole-image view with native windows, as HiDA-Net and GLASS do (Mu et al. 2026; Han 2026), is the obvious next step, but it would need its own held-out test. Of the two public detectors we tested, one is at chance on this data and the other does not benefit (Sec. 9), so we have not shown the effect outside our own training recipe.

13 Conclusion

A fixed-pixel window at native resolution fixes pixel count and leaves field of view free: the larger the image, the smaller the share the detector sees. For our DINOv2 detector on degraded images, that share limited robust AUC, not the representation and not a mismatch with the training view. Showing the whole image recovered 7 to 9 points per branch with no retraining, and training the full models on whole images added nothing more. In short proxy runs, CLIP and DINOv2 backbones both gained at test time (10 and 3 points) and more from training (4 and 5). On the test split's clean, large images the same change cost clean AUC, possibly the case the advice to avoid resizing was written for. B-Free, which follows that advice, lost clean AUC under every resized presentation and gained no robust AUC on the hard split. Detection papers should report the field of view their preprocessing implies, alongside resolution and crop policy.

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